

A Deep Learning Approach to Automate Classification of Arsenic-Affected Skin using EfficientNet-B1

Abstract—Arsenic contamination in groundwater poses a significant health risk, particularly in regions where it leads to chronic arsenic poisoning, manifesting in skin lesions and other dermatological conditions. Early and accurate detection of arsenic-affected skin is crucial for timely medical intervention. This study proposes an automated classification system using EfficientNet-B1, a state-of-the-art deep learning model, to detect arsenic-induced skin lesions from clinical images. The model was trained and evaluated on a dataset comprising images of healthy and arsenic-affected skin, with augmentation techniques employed to enhance robustness and generalizability from the dataset "ArsenicSkinImageBD: A comprehensive image dataset to classify affected and healthy skin of arsenic-affected people". For this classification task EfficientNet-B1 model was used and fine tuned for better performance. Which it delivered and the model achieved an accuracy of 92.76% Our results show that, EfficientNet-B1 has the potential to classify arsenic affected skin as a reliable and scalable tool for the automated diagnosis, which can aid healthcare providers in resource-limited settings. Future work will focus on expanding the dataset, enhancing model interpretability, and deploying the system in real-world clinical applications.

Index Terms—Deep Learning, EfficientNet-B1, Arsenic-Affected Skin, Automated Diagnosis, Medical Imaging

I. INTRODUCTION

Arsenic contamination in groundwater is a significant environmental and public health threat, notably in Bangladesh, India, and Southeast Asia, where millions of people are at danger of chronic arsenic poisoning. Excessive arsenic exposure can lead to skin sores, roughened skin (Hyperkeratosis), and increased risk of cancer. Early diagnosis of arsenic-induced skin problems is vital for prompt treatment and preventing subsequent health complications. Diagnosing arsenic-affected skin through clinical knowledge and visual inspection is subjective, time-consuming, and susceptible to human error [12]. Automated and objective procedures are essential for improving diagnosis accuracy and accessibility, especially in resource-limited situations.

Deep learning approach like Convolutional Neural Networks (CNNs) has showed potential in diagnosing skin cancer, diabetic retinopathy, and pneumonia using X-ray pictures. [18] [4] [5]. However, there is insufficient study on automatic classification of arsenic-affected skin. This work uses EfficientNet-B1, a cutting-edge deep learning model that balances accuracy and computational efficiency, making it ideal for deployment in clinical and low-resource applications.

This research utilizes the ArsenicSkinImageBD dataset, a comprehensive collection of high-resolution images depicting both healthy and arsenic-affected skin of individuals from arsenic-endemic regions [3]. The dataset offers a diverse set of images that capture various stages of arsenic-induced dermatological manifestations, making it an ideal resource for training robust classification models. By employing data augmentation techniques and transfer learning, the EfficientNet-B1 model is fine-tuned to accurately distinguish between healthy and affected skin, aiming to enhance diagnostic capabilities.

II. RESEARCH OBJECTIVE

A. Main Objective

This research seeks to find accuracy EfficientNet_B1 in classification of skin as infected or not infected by arsenic.

B. Research Question

This study aims to address the following research questions:

- 1) How efficiently does the EfficientNet-B1 model detect arsenic-affected skin lesions using the ArsenicSkinImageBD dataset?
- 2) How does the EfficientNet-B1 model compare to other deep learning models for classifying arsenic-affected skin?
- 3) Can the proposed model achieve a balance between accuracy and computational efficiency suitable for deployment in resource-limited settings?
- 4) How do data augmentation and transfer learning impact the model's ability to generalize to unseen arsenic-affected skin images?

III. RELATED WORKS

Deep learning and dermatology-an exciting intersection during recent years. It gets more interesting when skin diseases take into account those caused by even environmental toxins like arsenic. The prospect of classifying skin lesions that are affected by arsenic through an advanced neural network architecture such as EfficientNet-B1 offers a promising direction toward improving the diagnostic precision and efficiency. This section reviews the relevant literature underlining the importance of the deep learning methodology in the dermatological application related to arsenic-induced skin conditions.

Chronic exposure to arsenic is known to cause a range of skin disorders, including thickened skin (hyperkeratosis), darkened patches (hyperpigmentation), and serious conditions

like Bowen's disease and squamous cell carcinoma [8] [16]. The development of these conditions is complex, involving factors like oxidative stress and disruptions in the immune system, which have been widely reported in scientific studies [17]. For example, research by Huang et al. [7] highlights how arsenic exposure increases reactive oxygen species and triggers abnormal immune responses, which together contribute to the development of cancer. Understanding these mechanisms is essential for creating reliable diagnostic tools that can effectively identify and classify the different types of arsenic-induced skin lesions.

The application of deep learning in dermatology has shown remarkable promise, with studies demonstrating that convolutional neural networks (CNNs) can achieve diagnostic performance comparable to that of experienced dermatologists. Marchetti et al. trained a CNN on a large dataset of dermoscopic images, achieving dermatologist-level accuracy in skin cancer classification [11]. Similarly, Boffetta et al. conducted a systematic review that highlighted the effectiveness of machine learning algorithms in identifying skin lesions associated with arsenic exposure [1]. These findings suggest that deep learning models can be trained to recognize subtle features in skin lesions that may be indicative of arsenic toxicity. Afterwards, by early care and intervention these risks can be minimized.

Moreover, the integration of deep learning with image processing techniques has been explored to enhance the classification of skin lesions. Long et al. proposed a novel lightweight deep learning framework that leverages segmentation approaches to improve classification accuracy [6]. This methodology is particularly relevant for arsenic-affected skin lesions, as the visual characteristics of these lesions can vary significantly among individuals. The ability of deep learning models to adapt to these variations is crucial for developing robust diagnostic tools.

Besides the progress in deep learning models, genetic factors have also been explored in the context of arsenic-related skin issues. Research by Luo et al. looked at how genetic variations, known as gene polymorphisms, are linked to the development of skin lesions in people exposed to high levels of arsenic [10]. This study highlights the importance of considering genetic predispositions when designing machine learning models for classifying skin lesions. Including genetic data in these models could improve their accuracy and make diagnoses more tailored to individual patients, leading to more personalized healthcare.

Furthermore, the ethical considerations surrounding the collection of skin samples from arsenic-exposed individuals pose challenges for traditional diagnostic methods. Yang et al. highlighted the difficulties in obtaining skin samples for research purposes, which necessitates the development of non-invasive diagnostic techniques [16]. Deep learning approaches, particularly those utilizing image data, can circumvent these ethical issues by relying on readily available clinical images for training and validation.

The importance of comprehensive datasets in training deep learning models cannot be overstated. The performance of

these models is heavily dependent on the quality and diversity of the training data. Recent studies have emphasized the need for large, annotated datasets that encompass a wide range of skin lesions, including those caused by arsenic exposure [2] [15]. By utilizing diverse datasets, researchers can ensure that their models are capable of generalizing across different populations and skin types, thereby improving diagnostic accuracy.

In conclusion, the integration of deep learning techniques in the classification of arsenic-affected skin lesions holds significant promise for advancing dermatological diagnostics. The existing literature provides a robust foundation for understanding the complexities of arsenic-induced skin disorders and the potential of machine learning to enhance diagnostic accuracy. Future research should focus on expanding datasets, incorporating genetic factors, and addressing ethical challenges to fully realize the potential of deep learning in this critical area of public health.

IV. METHODOLOGY

This section outlines the methodology employed in the development and evaluation of the proposed automated classification system for arsenic-affected skin using the EfficientNet-B1 model. It is structured into several key phases: dataset preparation, model architecture selection, training and tuning, and evaluation of performance metrics. Each phase is crucial for ensuring the robustness and accuracy of the classification model.

A. Data Preparation

The study utilized the **ArsenicSkinImageBD** dataset, which contains a total of 8,892 skin images. The dataset is evenly split, with half images depicting skin affected by arsenic exposure and half images showing mixed skin conditions that are not affected by arsenic.



Fig. 1. The palm of an affected person on the left and the palm of a healthy person on the right.

This balanced dataset is essential for mitigating bias during the training process and ensuring that the model learns to differentiate effectively between the two classes. The images are diverse, capturing various stages and types of arsenic-induced dermatological manifestations, along with healthy and other non-arsenic-related skin conditions.

1) *Data Preprocessing*: Before training the model, the images were resized to 240×240 pixels to fit the input requirements of the EfficientNet-B1 architecture. Image normalization was performed to scale pixel values between 0 and 1, enhancing the training stability.



Fig. 2. The skin of an affected person on the left and the skin of a healthy person on the right.

2) *Data Augmentation*: : To improve the model's robustness and generalizability, data augmentation techniques such as random rotations, flips, shifts, and zooms were applied to the dataset. This approach helps to simulate variability in real-world clinical settings and prevents overfitting.

3) *Dataset Split*: : The dataset was split into two subsets: 70% for training, and 30% for testing. The split was performed randomly to ensure a balanced representation of arsenic-affected and healthy skin images in each subset.

B. Model Architecture Selection

The EfficientNet-B1 architecture was chosen for this study due to its state-of-the-art performance in image classification tasks, particularly in the medical domain [13]. EfficientNet models are known for their efficiency and accuracy, achieved through a compound scaling method that uniformly scales the depth, width, and resolution of the network. EfficientNet uses a compound scaling method that balances the network's depth, width, and resolution. This allows for better performance with fewer parameters compared to traditional CNN architectures.

$$\text{depth} = \alpha^d, \quad \text{width} = \beta^d, \quad \text{resolution} = \gamma^d \quad (1)$$

$$\alpha \times \beta^2 \times \gamma^2 \approx 2 \quad (2)$$

C. Training and Tuning

The training process involved splitting the dataset into training, validation, and test sets, with 70% of the data allocated for training and 15% for testing. The training phase utilized a batch size of 32 with 20 epoch and employed the Adam optimizer, which is known for its efficiency in handling sparse gradients and adapting learning rates.

During training, the model underwent several epochs, with early stopping implemented to prevent overfitting. The learning rate was adjusted dynamically based on the validation loss, allowing for fine-tuning of the model's parameters. The loss function used was categorical cross-entropy, suitable for the classification problem at hand.

D. Evaluation Metrics

After training, the model's performance was evaluated using the test set, with accuracy being the primary metric of interest. The EfficientNet-B1 model achieved an impressive accuracy of 92.76%, indicating its effectiveness in distinguishing between arsenic-affected skin lesions and non-affected skin images.

In addition to accuracy, other performance metrics such as precision, recall, and F1-score were calculated to provide a comprehensive evaluation of the model's performance. These metrics are particularly important in medical image classification, where the cost of false negatives can be significant. A confusion matrix was also generated to visualize the model's performance across the two classes, providing insights into areas where the model may need improvement.

E. Implementation Details

The model was implemented using Python with the TensorFlow and Keras libraries on an NVIDIA GPU T4 -enabled environment, which significantly accelerated the training process. The methodology employed in this study effectively harnessed the capabilities of deep learning, particularly the EfficientNet-B1 architecture, to automate the classification of arsenic-affected skin lesions. The combination of a well-prepared dataset, robust model architecture, and thorough training and evaluation processes contributed to the high accuracy achieved, highlighting the potential of deep learning in dermatological diagnostics.

V. RESULTS

This section presents the performance evaluation of the EfficientNet-B1 model in classifying arsenic-affected and non-affected skin images using the ArsenicSkinImageBD dataset. The evaluation metrics used include accuracy, precision, recall, F1 score, the ROC curve, and the confusion matrix.

TABLE I
PERFORMANCE METRICS OF THE EFFICIENTNET-B1 MODEL

Metric	Value
Accuracy	0.9276
Precision	0.9577
Recall	0.8896
F1 Score	0.9224
AUC (Area Under the Curve)	0.98
True Positives (TP)	1520
False Positives (FP)	58
True Negatives (TN)	1313
False Negatives (FN)	163

A. Performance Metrics

The classification performance of the EfficientNet-B1 model was evaluated using the following metrics:

1) *Accuracy*: : The model achieved an overall accuracy of 92.76%, which indicates that it correctly classified 92.76% of the test images. This high accuracy demonstrates the model's capability to effectively distinguish between arsenic-affected and non-affected skin.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

2) *Precision*: : The model achieved a precision score of 95.77%, indicating that out of all the images predicted to show arsenic-affected skin, 95.77% were correctly classified. This suggests the model is highly effective at minimizing false positives.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

3) *Recall*: : The recall score was 88.96%, meaning the model correctly identified 88.96% of all arsenic-affected skin images. Although slightly lower than precision, this recall value indicates that the model performs well at detecting true positive cases.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

4) *F1 Score*: : The F1 score, which balances precision and recall, was 92.23%. This value reflects the model's overall performance and suggests a good balance between minimizing false positives and false negatives.

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

B. Confusion Matrix

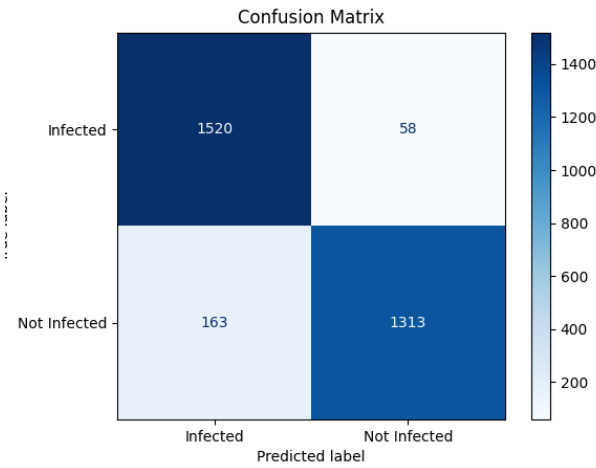


Fig. 3. Confusion Matrix

The confusion matrix, as shown in Figure 1, provides a detailed breakdown of the model's predictions:

1) *True Positives (TP)*: : 1,520 images of arsenic-affected skin were correctly classified as "Infected."

2) *True Negatives (TN)*: : 1,313 images of non-affected skin were correctly classified as "Not Infected."

3) *False Positives (FP)*: : 58 images were incorrectly classified as "Infected" when they were actually "Not Infected."

4) *False Negatives (FN)*: : 163 images were incorrectly classified as "Not Infected" when they were actually "Infected."

C. ROC Curve and AUC Analysis

The Receiver Operating Characteristic (ROC) curve, depicted above, illustrates the trade-off between the true positive rate (sensitivity) and the false positive rate across different threshold settings. The area under the curve (AUC) is 0.98, indicating excellent model performance in distinguishing between arsenic-affected and non-affected skin images. The formula used to calculate AUC is:

$$\text{AUC} = \int_0^1 \text{TPR}(t) d(\text{FPR}(t)) \quad (7)$$

An AUC of 0.98 confirms that the model has a high discriminative ability, correctly classifying cases with high probability. This measure is particularly valuable in understanding the model's capability to maintain performance across varying classification thresholds, affirming its reliability in clinical decision-making scenarios.

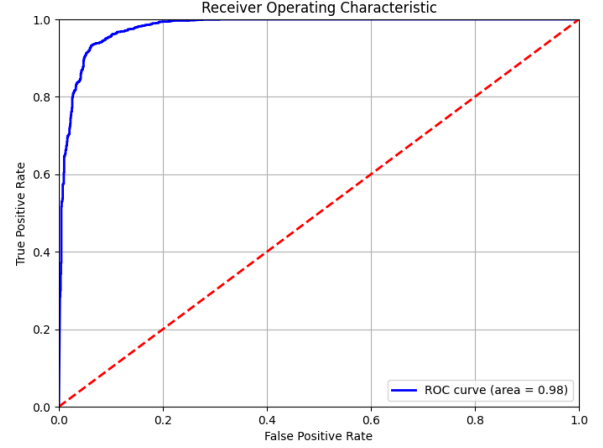


Fig. 4. ROC-Curve

The confusion matrix reveals that the model has a strong ability to correctly classify most images, though there is a slight imbalance with more false negatives than false positives. This indicates that while the model is effective, further tuning could be done to improve the recall score, ensuring that fewer arsenic-affected cases go undetected.

D. Comparative Analysis

The performance metrics achieved by the EfficientNet-B1 model, including an accuracy of 92.76% and an F1 score of 92.23%, compare favorably with other deep learning models in similar image classification tasks [9]. While the model achieves high precision and accuracy, the slightly lower recall suggests that there is room for improvement in detecting all true positive cases. Future work could explore techniques like adjusting class weights or implementing additional data augmentation strategies to improve recall without compromising precision.

E. Implications and Limitations

The high performance of the EfficientNet-B1 model suggests its potential for clinical deployment as a diagnostic aid for identifying arsenic-affected skin lesions. However, the presence of false negatives highlights a need for caution, as missing affected cases could delay necessary interventions. Future research could explore additional data augmentation techniques, alternative model architectures, or integration with clinical data to further improve recall and overall diagnostic accuracy.

VI. DISCUSSIONS

The purpose of this study was to develop a deep learning-based automated classification system for identifying arsenic-affected skin lesions using the EfficientNet-B1 model. The results demonstrate that the proposed model performs effectively, achieving an accuracy of 92.76%, precision of 95.77%, recall of 88.96%, and an F1 score of 92.23%. These metrics underscore the model's potential as a reliable tool for assisting healthcare professionals in diagnosing arsenic-related dermatological conditions.

A. Interpretation of Results

The high accuracy and precision of the EfficientNet-B1 model indicate its robustness in distinguishing between arsenic-affected and non-affected skin images. The model's performance is further validated by an AUC of 0.98, suggesting excellent discriminative ability. The confusion matrix analysis shows a relatively low number of false positives, which is crucial in a clinical context as it reduces the risk of misdiagnosing healthy skin as arsenic-affected.

However, the model's recall (88.96%) is slightly lower than its precision, indicating that it occasionally fails to identify arsenic-affected cases, leading to false negatives. This suggests that the model, while highly precise, may miss some true cases of arsenic-related skin conditions. In a healthcare setting, such false negatives could delay the necessary medical intervention for affected individuals, highlighting an area for improvement.

B. Implications of the Study

The successful implementation of the EfficientNet-B1 model has significant implications for clinical practice, particularly in areas with high arsenic exposure. Automated skin classification systems can provide quick, cost-effective, and reliable preliminary diagnoses, especially in rural and resource-limited settings where access to trained dermatologists is scarce. This technology could be integrated into mobile applications, allowing individuals to self-screen for arsenic-affected skin conditions and seek medical advice if necessary.

Additionally, the deployment of such models can alleviate the burden on healthcare professionals by acting as a decision-support tool, ensuring that more patients receive timely assessments. This approach aligns with broader public health initiatives aimed at reducing the impact of environmental health hazards such as arsenic exposure.

C. Limitations

Despite the promising results, the study has some limitations that should be considered:

1) *Dataset Constraints*: : The model was trained and evaluated using the ArsenicSkinImageBD dataset, which, although comprehensive, may not fully capture the global variability in arsenic-affected skin conditions. Variations in lighting, image quality, and skin types across different populations could affect the model's generalizability.

2) *Lack of External Validation*: : The model was not tested on external datasets, which limits the ability to validate its performance in different real-world settings. Future studies should involve cross-dataset validation to assess the robustness and adaptability of the model across diverse populations.

3) *False Negatives*: : The model's relatively lower recall suggests that some cases of arsenic-affected skin were not detected. This limitation could impact the clinical applicability of the model, as missing true positive cases might delay critical medical interventions.

D. Future Directions

To address these limitations and enhance the model's performance, several future research directions can be explored:

1) *Expanding the Dataset*: : Incorporating additional images from diverse populations and environmental conditions could improve the model's ability to generalize across different settings. Data augmentation techniques that simulate real-world variability could also be beneficial.

2) *Improving Recall*: : Techniques such as adjusting the decision threshold, employing ensemble models, or fine-tuning class weights could be explored to improve recall without significantly affecting precision.

3) *Explainability and Interpretability*: : Future work should focus on making the model's decision-making process more interpretable, especially for clinical use. Techniques such as Grad-CAM (Gradient-weighted Class Activation Mapping) could help visualize which parts of the image the model focuses on, enhancing trust and usability among healthcare professionals.

4) *Integration with Clinical Data*: : Combining image data with other clinical information, such as patient history and symptoms, could provide a more holistic approach to diagnosing arsenic-affected skin conditions, potentially increasing the overall accuracy and reliability of the system.

VII. CONCLUSION

This study demonstrates the effectiveness of using the EfficientNet-B1 deep learning model to automate the classification of arsenic-affected skin lesions. By leveraging the ArsenicSkinImageBD dataset, which consists of 8,892 diverse skin images, the model was trained to distinguish between arsenic-affected and healthy skin with an impressive accuracy of 92.76.

The implications of this research extend beyond mere classification; the automated system can serve as a diagnostic tool that aids healthcare professionals in identifying arsenic-induced skin conditions, potentially improving patient outcomes through early intervention. Furthermore, the findings emphasize the importance of integrating advanced machine learning techniques in dermatological diagnosis, paving the way for future research in this domain [14].

However, the study is not without limitations. The reliance on a single dataset may restrict the generalizability of the findings, and further validation with larger and more diverse datasets is necessary to confirm the model's performance.

Future work should also explore the interpretability of the model's predictions and the potential integration of additional clinical data to enhance diagnostic accuracy.

In conclusion, this research contributes to the growing body of evidence supporting the use of deep learning in medical image analysis, specifically for conditions arising from environmental exposures like arsenic. As technology continues to advance, the potential for automated systems in healthcare promises to revolutionize diagnostic practices and improve access to medical care for affected populations.

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